EX 07 :

1.Create Database and Collection

use payrollDB

db.createCollection("employees")

```
> use payrollDB
< switched to db payrollDB
> db.createCollection("employees")
< { ok: 1 }
```

2. Insert Sample Data

db.employees.insertMany([

{ emp_id: "E101", name: "John", department: "HR", basic_salary: 45000, hra: 8000, allowances: 3000 },

{ emp_id: "E102", name: "Mary", department: "IT", basic_salary: 60000, hra: 10000, allowances: 5000 },

{ emp_id: "E103", name: "Sam", department: "Finance", basic_salary: 55000, hra: 9000, allowances: 4000 },

{ emp_id: "E104", name: "David", department: "IT", basic_salary: 65000, hra: 11000, allowances: 7000 },

{ emp_id: "E105", name: "Rose", department: "HR", basic_salary: 48000, hra: 8500, allowances: 3500 }

])

```
> db.employees.insertMany([
    { emp_id: "E101", name: "John", department: "HR", basic_salary: 45000, hra: 8000, allowances: 3000 },
    { emp_id: "E102", name: "Mary", department: "IT", basic_salary: 60000, hra: 10000, allowances: 5000 },
    { emp_id: "E103", name: "Sam", department: "Finance", basic_salary: 55000, hra: 9000, allowances: 4000 },
    { emp_id: "E104", name: "David", department: "IT", basic_salary: 65000, hra: 11000, allowances: 7000 },
    { emp_id: "E105", name: "Rose", department: "HR", basic_salary: 48000, hra: 8500, allowances: 3500 }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69bd5b6c0bcb88b0c2d41c32'),
      '1': ObjectId('69bd5b6c0bcb88b0c2d41c33'),
      '2': ObjectId('69bd5b6c0bcb88b0c2d41c34'),
      '3': ObjectId('69bd5b6c0bcb88b0c2d41c35'),
      '4': ObjectId('69bd5b6c0bcb88b0c2d41c36')
    }
  }
```

3.Apply Aggregate Queries

i) Total Salary of Each Employee

db.employees.aggregate([{ $project: { name: 1, department: 1, total_salary: { $add: ["$basic_salary", "$hra", "$allowances"] } } }])

```
  _id: ObjectId('69bd5b6c0bcb88b0c2d41c33'),
  name: 'Mary',
  department: 'IT',
  total_salary: 75000
}
{
  _id: ObjectId('69bd5b6c0bcb88b0c2d41c34'),
  name: 'Sam',
  department: 'Finance',
  total_salary: 68000
}
{
  _id: ObjectId('69bd5b6c0bcb88b0c2d41c35'),
  name: 'David',
  department: 'IT',
  total_salary: 83000
}
{
  _id: ObjectId('69bd5b6c0bcb88b0c2d41c36'),
  name: 'Rose',
  department: 'HR',
  total_salary: 60000
}
```

ii) Total Salary Paid by Organization

db.employees.aggregate([{ $group: { _id: null, total_paid: { $sum: { $add: ["$basic_salary", "$hra", "$allowances"] } } } }])

```
{
  _id: null,
  total_paid: 342000
}
```

iii) Department-wise Average Salary

db.employees.aggregate([{ $group: { _id: "$department", avg_salary: { $avg: "$basic_salary" } } }])

```
{
  _id: 'IT',
  avg_salary: 62500
}
{
  _id: 'HR',
  avg_salary: 46500
}
{
  _id: 'Finance',
  avg_salary: 55000
}
```

iv) Highest & Lowest Salary in Each Department

db.employees.aggregate([{ $group: { _id: "$department", max: { $max: "$basic_salary" }, min: { $min: "$basic_salary" } } }])

```
{
  _id: 'Finance',
  max: 55000,
  min: 55000
}
{
  _id: 'HR',
  max: 48000,
  min: 45000
}
{
  _id: 'IT',
  max: 65000,
  min: 60000
}
```

v) Count Employees in Each Department

db.employees.aggregate([{ $group: { _id: "$department", emp_count: { $sum: 1 } } }])

```
{
  _id: 'Finance',
  emp_count: 1
}
{
  _id: 'HR',
  emp_count: 2
}
{
  _id: 'IT',
  emp_count: 2
}
```

vi) Sort Employees by Salary (Descending)

db.employees.aggregate([{ $sort: { basic_salary: -1 } }])

```
< {
    _id: ObjectId('69bd5b6c0bcb88b0c2d41c35'),
    emp_id: 'E104',
    name: 'David',
    department: 'IT',
    basic_salary: 65000,
    hra: 11000,
    allowances: 7000
  }
  {
    _id: ObjectId('69bd5b6c0bcb88b0c2d41c33'),
    emp_id: 'E102',
    name: 'Mary',
    department: 'IT',
    basic_salary: 60000,
    hra: 10000,
    allowances: 5000
  }
```

```
{
  _id: ObjectId('69bd5b6c0bcb88b0c2d41c34'),
  emp_id: 'E103',
  name: 'Sam',
  department: 'Finance',
  basic_salary: 55000,
  hra: 9000,
  allowances: 4000
}
{
  _id: ObjectId('69bd5b6c0bcb88b0c2d41c36'),
  emp_id: 'E105',
  name: 'Rose',
  department: 'HR',
  basic_salary: 48000,
  hra: 8500,
  allowances: 3500
}
```

```
{
  _id: ObjectId('69bd5b6c0bcb88b0c2d41c32'),
  emp_id: 'E101',
  name: 'John',
  department: 'HR',
  basic_salary: 45000,
  hra: 8000,
  allowances: 3000
}
```